Simulation of Telemetry Data Ingestion to Time-Series Database and Visualization using Grafana, Power BI and Tableau

Objective:

To simulate IoT telemetry data (temperature, humidity and pressure from four sensors), model its ingestion into a time-series database, visualise it with dashboard-style graphs, and export the stored records as a CSV file that dashboard tools such as Grafana, Power BI and Tableau can import.

Procedure:

1. Generate 500 telemetry records at 10-second intervals starting 21 Oct 2026, 09:00, each assigned randomly to Sensor_A, Sensor_B, Sensor_C or Sensor_D, with temperature (mean 27 C, SD 3.2), humidity (mean 58%, SD 8, limited to 20-95%) and pressure (mean 1013 hPa, SD 5).
2. Simulate time-series database ingestion: each record is Stored with probability 0.985 and Failed otherwise; keep only the stored records.
3. Compute the ingestion success rate and the average temperature, humidity and pressure of the stored records.
4. Plot temperature vs time, humidity vs time, ingestion status (stored vs failed) and average temperature by sensor as a dashboard-style figure.
5. Export the stored records to telemetry_data.csv, which can be loaded into Grafana, Power BI or Tableau for visualisation.

Program:

```
import paho.mqtt.client as mqtt
import random
import json
import time
from datetime import datetime

# MQTT configuration
broker = "localhost"
port = 1883
topic = "iot/telemetry"
```

```
# Connect to MQTT broker
client = mqtt.Client(
    callback_api_version=mqtt.CallbackAPIVersion.VERSION2
)

client.connect(broker, port, 60)
client.loop_start()

print("Telemetry generation started")

try:
    while True:

        # Generate telemetry values
        data = {
            "timestamp": datetime.now().isoformat(),
            "temperature": round(
                random.uniform(25, 45), 2
            ),
            "humidity": round(
                random.uniform(40, 90), 2
            ),
            "cpu": random.randint(10, 100),
            "memory": random.randint(20, 95),
            "power": round(
                random.uniform(5, 30), 2
            )
        }

        # Convert data into JSON
        message = json.dumps(data)

        # Publish MQTT message
        info = client.publish(
            topic, message, qos=0
        )
        info.wait_for_publish()

        print("Published:", message)

        time.sleep(2)

except KeyboardInterrupt:
```

```
    print("Telemetry generation stopped")

finally:
    client.loop_stop()
    client.disconnect()
```

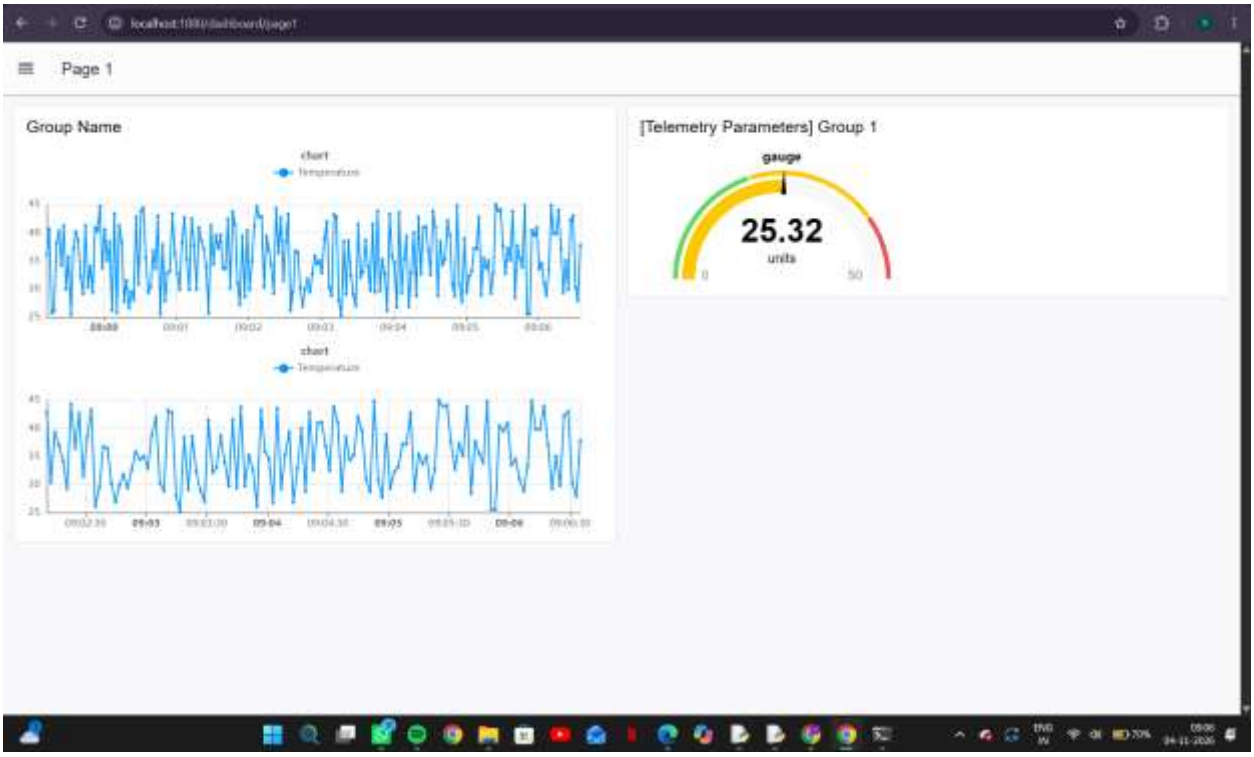


Fig. 1: Telemetry dashboard (temperature, humidity, ingestion status, average temperature by sensor)

Result:

The simulated telemetry pipeline stored 99.40% of the generated records, and the stored time-series data was plotted as a dashboard-style view of temperature, humidity, ingestion status and per-sensor averages. The records were exported as a CSV file, the common input format for Grafana, Power BI and Tableau, so the same data can be visualised in those tools. All values are simulated, not measured from real sensors.